AAEC4484/AAEC(STAT)5484: Applied Economic Forecasting

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Introduction

In this guide, we will access and analyze agricultural data from the USDA National Agricultural Statistics Service (NASS) using the `tidyUSDA` package in `R`. The USDA NASS provides a wealth of data on crop production, livestock, and other agricultural statistics, which can be invaluable for economic forecasting and analysis.

Accessing USDA NASS Data

To access USDA NASS data, you will need to obtain an API key from the USDA. You can sign up for an API key at https://quickstats.nass.usda.gov/api. Once you have your API key, you can securely store it using the `keyring` package in `R`. Now, you will be able to share code publicly without exposing your API key.

Install the `keyring` package if you haven't already.

```
# Store your API key securely using the key_set function
#tidyUSDA is the nickname assigned for my API key combination
keyring::key_set("tidyUSDA", "your_api_key_here")
```

A password prompt should now appear. Enter your API key and click `OK`.

Now, you can access the API key in your code later on using the `key_get()` function– removing the need for hardcoding your key.

```
key_get("tidyUSDA")
```

If you forgot the name of the service you used to store your API key, you can list all stored keys with the `key_list()` function.

```
key_list()
```

Fetching data from USDA NASS

Once you have your API stored, we can now use the `tidyUSDA` package to fetch data from the USDA NASS. For example, let's say we want to retrieve data on national corn prices received in 1990 to 2024.

We can do so with the `getQuickstat()` function, which allows us to specify the parameters for our data query. Below is an example of how to fetch this data.

```
library(tidyUSDA)
# Fetch corn price data for Iowa from 1990 to 2025
corn_prices <- tidyUSDA::getQuickstat(
  key = keyring::key_get("tidyUSDA"),
  program = "SURVEY",
  sector = "CROPS",
  group = "FIELD CROPS",
  commodity = "CORN",
  category = "PRICE RECEIVED",
  data_item = "CORN, GRAIN - PRICE RECEIVED, MEASURED in $ / BU",
  domain = "TOTAL",
  geographic_level = "NATIONAL",
  state = "US TOTAL",
#Must specify years as character variables
```

```
  year = as.character(c(1990:2024))
)
```

If you want to better understand the API query, you can visit the [USDA NASS Quick Stats Website https://quickstats.nass.usda.gov/ and see the matching arguments you used in the `getQuickstat()` function.

Advantage

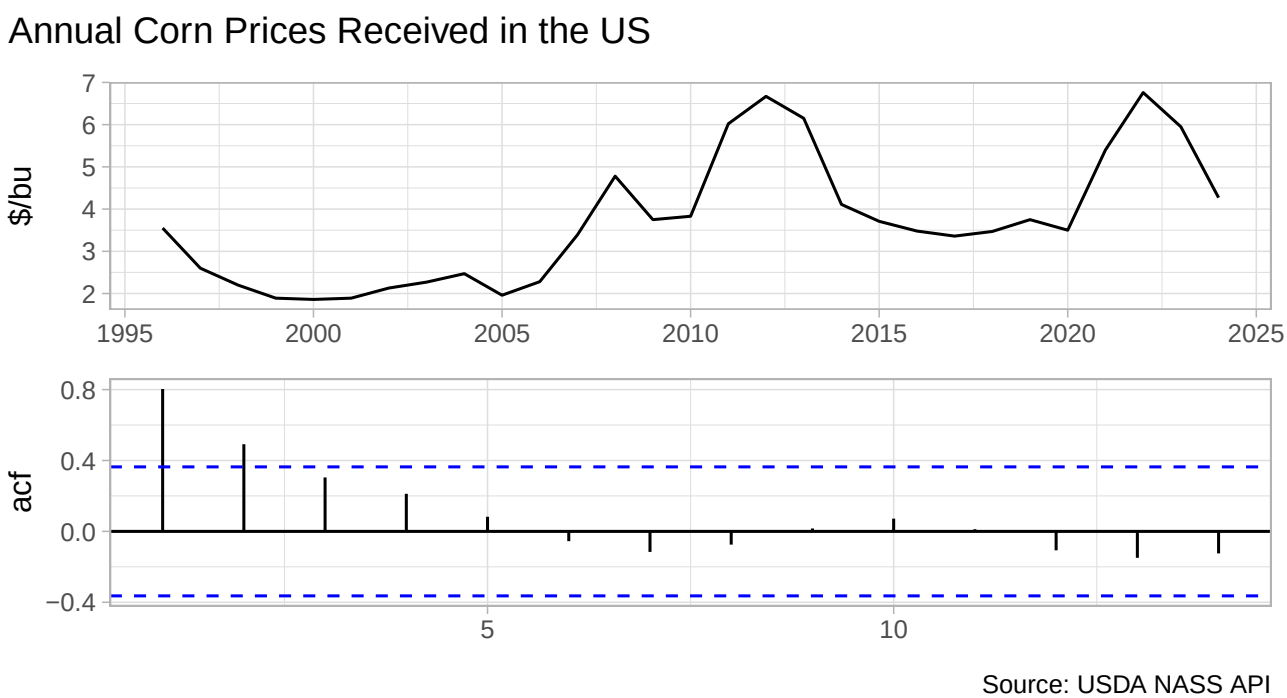
The API will save us a lot of time and effort in data collection. We can retrieve data for multiple commodities, states, and years with just a few lines of code and update our data easily in the future by changing the parameters in our query.

Analyzing the Data

Since we would like to use the functionalities from the `fpp3` package, we need to convert our data into a `tsibble` format.

For this exercise, we are interested in the Annual values so we will filter on the `reference_period_desc` column and report only the prices.

```
prices <- corn_prices %>%
  filter(reference_period_desc == "YEAR") %>%
  select(year, Value) %>%  as_tsibble(index = year)
```

Now that we have our data in the correct format, we can proceed as normal.

```
a <- prices %>% autoplot(Value) + labs(y = "$/bu", x = NULL)
b <- prices %>% ACF(Value) %>% autoplot() + labs(x = NULL)

a/b  + plot_annotation(title = "Annual Corn Prices Received in the US",
                       caption = "Source: USDA NASS API")
```